

Saif ur Rehman

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Professional Summary

AI Engineer with experience designing and building autonomous agents and LLM-driven workflows for scalable backend services. Skilled in prompt chaining, microservices, and serverless architectures on AWS, with expertise in Node.js, Python, and CI/CD pipelines. Demonstrated ability to automate development workflows and optimize performance for production AI applications. Adept at delivering high-quality, compliant AI solutions that enhance developer productivity.

Education

FAST National University of Computer and Emerging Sciences

Aug 2022 – Jun 2026

Bachelor of Science, Computer Science

Islamabad, Pakistan

- Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Data Structures, Database Systems, Cloud Computing, Distributed Systems

Technical Skills

AI & ML: LLM Integration (OpenAI, Anthropic, Hugging Face), RAG Systems, Prompt Engineering, Agentic Workflows, Fine-Tuning, Semantic Search, Vector Databases (Pinecone, Weaviate)

Agent Frameworks: LangChain, LangGraph, AutoGPT, CrewAI, Multi-Agent Orchestration, Tool Calling, Memory Management

Backend Development: Python (FastAPI), Node.js, REST APIs, Microservices, Asynchronous Programming, WebSocket, API Integration, MERN Stack

Cloud & DevOps: AWS (Lambda, EC2, S3, DynamoDB, Step Functions), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions)

Databases: PostgreSQL, MongoDB, DynamoDB, Vector Databases, Query Optimization

Tools & Practices: Git, Linux, Postman, PyTorch, Hugging Face Diffusers, Agile Development, System Design, Jira, n8n, Autonomous Workflows

Experience

Nexium

Jun 2025 – Aug 2025

Software Engineering Intern

Lahore, Pakistan

- Engineered scalable backend services and RESTful APIs for AI-enabled applications, implementing prompt chaining and LLM integration workflows.
- Developed and deployed autonomous data processing workflows using n8n and serverless architecture on AWS, reducing manual workload by 60%.
- Built event-driven MERN stack microservices with Node.js and React integrations that used LLM APIs for intelligent decision-making and automated response generation.
- Optimized API response times by 40% through efficient asynchronous programming, database query optimization, and automated deployment via GitHub Actions CI/CD pipelines.
- Implemented comprehensive monitoring and logging systems with issue tracking in Jira for real-time AI agent performance tracking and observability.
- Collaborated in agile sprints to deliver production-ready features, with a focus on reliability and user experience.

Projects

Autonomous AI Agent for Intelligent Document Processing | LangChain, FastAPI, Docker, Kubernetes

2025

- Designed and implemented a multi-agent orchestration framework using LangChain for complex document processing workflows.
- Built RAG pipelines with semantic search capabilities, leveraging vector databases for efficient context retrieval and response generation.

- Developed sophisticated prompt engineering strategies including chain-of-thought prompting and tool calling for enhanced agent reasoning.
- Integrated LLM APIs (OpenAI GPT-4, Anthropic Claude) with custom fine-tuning for domain-specific accuracy improvements.
- Deployed the agent system using FastAPI in Docker containers orchestrated with Kubernetes, managing service definitions in YAML for environment-agnostic deployments.
- Created CI/CD pipelines with GitHub Actions using YAML definitions to automate testing, builds, and deployments.

AI-Powered Animated Video Generation System | *LangChain, LangGraph, ElevenLabs, Stable Diffusion, FFmpeg, FastAPI* 2025

- Architected an end-to-end multi-agent pipeline that accepts a natural language prompt and autonomously produces a complete short animated video – covering story generation, voice synthesis, visual composition, and final MP4 export – with zero manual creative intervention.
- Orchestrated agent-to-agent data flow through a shared JSON state schema, enabling modular handoff across LLM story generation (LangChain), TTS audio synthesis (ElevenLabs/Bark), image generation (Stable Diffusion/DALL-E), and video compositing (FFmpeg/MoviePy).
- Implemented an intelligent editing agent using LangGraph with intent classification, targeted component re-execution, and full state versioning with undo/revert support via SQLite snapshots.

Cloud-Native Microservices with AI Integration | *AWS EKS, Terraform, GitHub Actions, Step Functions* 2024

- Architected and deployed containerized microservices on AWS EKS with integrated AI processing capabilities.
- Implemented Infrastructure as Code using Terraform for reproducible and scalable AI infrastructure deployment.
- Built CI/CD pipelines with GitHub Actions for automated testing and deployment of AI services.
- Designed event-driven architecture using AWS Step Functions for orchestrating complex AI workflows.

LabGuard – Monitoring & Compliance Analytics System | *Node.js, MongoDB, WebSocket, REST APIs* 2024

- Developed real-time monitoring platform with WebSocket integration for instant event tracking and alerting.
- Built high-throughput event processing pipelines capable of handling thousands of concurrent operations.
- Implemented analytics dashboards with automated reporting and anomaly detection capabilities.
- Designed RESTful APIs for seamless integration with frontend visualization and third-party services.

Generative AI Research – Diffusion Models | *PyTorch, Hugging Face Diffusers, DDPM, Latent Diffusion* 2025

- Conducted comparative analysis of DDPM and Latent Diffusion architectures, evaluating performance on domain-specific datasets.
- Analyzed model outputs using FID and Inception Score metrics to optimize generation quality and stability.
- Explored optimization techniques for efficient model training and inference pipelines.
- Developed custom evaluation frameworks for assessing model performance in production scenarios.

Certifications

Teaching Assistant: Operating Systems, FAST National University (Mentored 100+ students in system-level programming)

AWS Cloud Foundations: AWS Academy (Certified in cloud infrastructure and serverless computing)

MERN Stack Certification: freeCodeCamp (Full-stack web development)